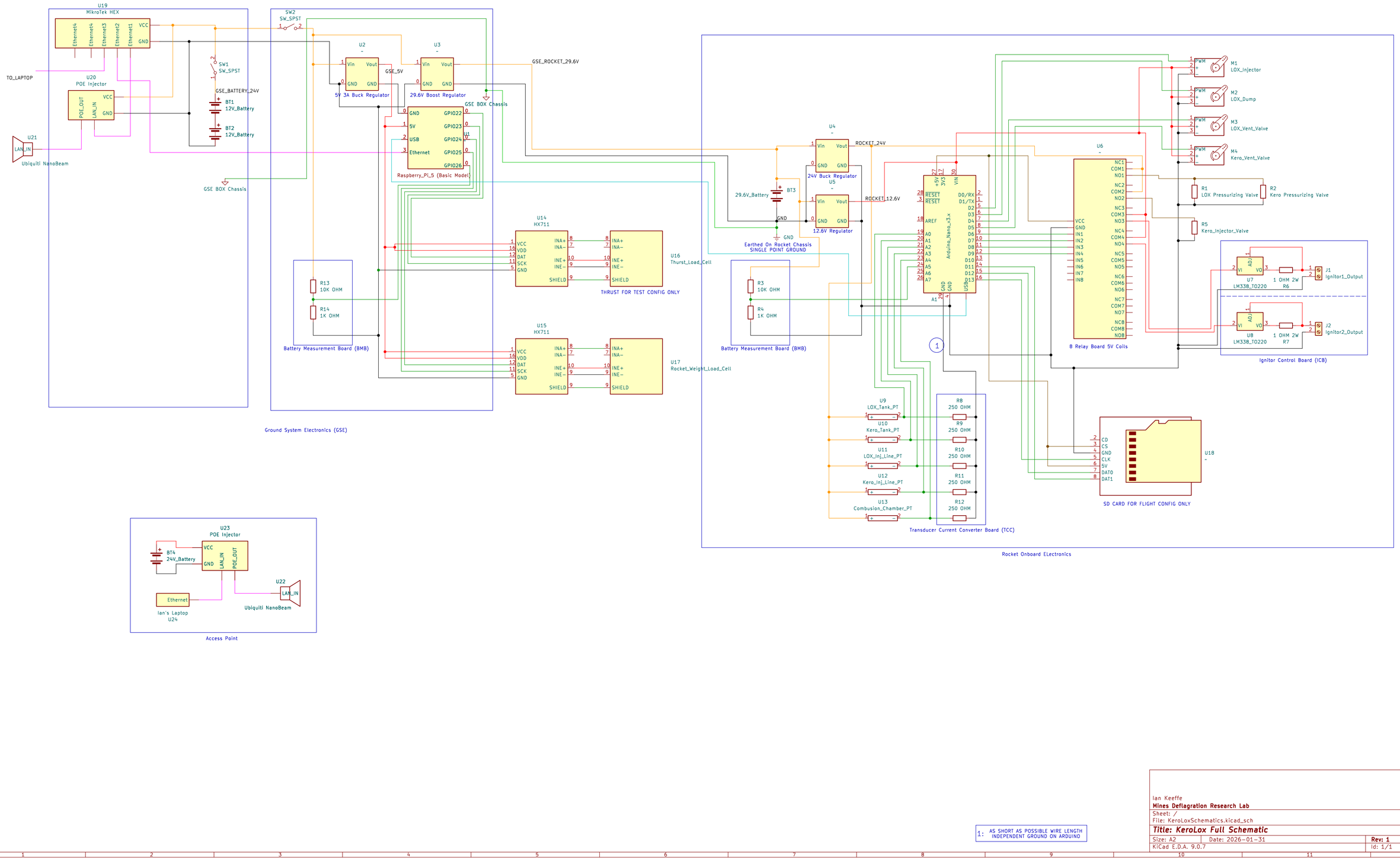

U19
MikroTek HEX
Ethernet4
Ethernet4
Ethernet3
Ethernet2
Ethernet1
VCC
GND
TO_LAPTOP
U20
POE Injector
POE_OUT
LAN_IN
VCC
GND
U21
LAN_IN
Ubiquiti NanoBeam
SW1
SW_SPST
GSE_BATTERY_24V
BT1
12V_Battery
BT2
12V_Battery
GSE BOX Chassis
SW2
SW_SPST
U2
5V 3A Buck Regulator
U3
29.6V Boost Regulator
Vin
Vout
GND
GSE_5V
GSE_ROCKET_29.6V
GSE BOX Chassis
Raspberry_Pi_5 (Basic Model)
GND
5V
USB
Ethernet
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26
U14
HX711
U15
HX711
U16
Thurst_Load_Cell
U17
Rocket_Weight_Load_Cell
THRUST FOR TEST CONFIG ONLY
R13
10K OHM
R14
1K OHM
Battery Measurement Board (BMB)
Ground System Electronics (GSE)
U4
24V Buck Regulator
U5
12.6V Regulator
ROCKET_24V
ROCKET_12.6V
29.6V_Battery
BT3
GND
Earthed On Rocket Chassis
SINGLE POINT GROUND
R3
10K OHM
R4
1K OHM
Battery Measurement Board (BMB)
Arduino_Nano_v3.x
U6
8 Relay Board 5V Coils
M1
LOX_Injector
M2
LOX_Dump
M3
LOX_Vent_Valve
M4
Kero_Vent_Valve
R1
LOX Pressurizing Valve
R2
Kero Pressurizing Valve
R5
Kero_Injector_Valve
U7
LM338_TO220
1 OHM 2W
R6
J1
Ignitor1_Output
U8
LM338_TO220
1 OHM 2W
R7
J2
Ignitor2_Output
Ignitor Control Board (ICB)
U9
LOX_Tank_PT
U10
Kero_Tank_PT
U11
LOX_Inj_Line_PT
U12
Kero_Inj_Line_PT
U13
Combusion_Chamber_PT
R8
250 OHM
R9
250 OHM
R10
250 OHM
R11
250 OHM
R12
250 OHM
Transducer Current Converter Board (TCC)
U18
CD
CS
GND
CLK
5V
DAT0
DAT1
SD CARD FOR FLIGHT CONFIG ONLY
Rocket Onboard ELectronics
U23
POE Injector
BT4
24V_Battery
VCC
GND
LAN_IN
POE_OUT
U22
LAN_IN
Ethernet
Ian's Laptop
U24
Ubiquiti NanoBeam
Access Point
1: AS SHORT AS POSSIBLE WIRE LENGTH INDEPENDENT GROUND ON ARDUINO
Ian Keeffe
Mines Deflagration Research Lab
Sheet: /
File: KeroLoxSchematics.kicad_sch
Title: KeroLox Full Schematic
Size: A2
Date: 2026-01-31
Rev: 1
KiCad E.D.A. 9.0.7
Id: 1/1